

BookNest — PHP + MySQL Bookstore Project

A complete college-level e-commerce project with a PHP backend, MySQL database, and a single-page frontend.

Project Structure

```
booknest/
├── index.php           ← Main frontend (single-page app)
├── styles.css          ← Separated CSS styling
├── database.sql        ← Run once to set up DB + seed data
├── includes/
│   └── config.php     ← DB credentials & helper functions
└── api/
    ├── auth.php       ← Login / Register / Logout / Session
    ├── books.php      ← Book listing, search, filter, sort
    ├── cart.php       ← Cart (DB for users, Session for guests)
    ├── orders.php     ← Place order, order history
    └── contact.php    ← Contact form submissions
```

Setup Instructions

1. Requirements

- PHP 7.4 or higher (PHP 8.x recommended)
- MySQL 5.7+ or MariaDB 10+
- Apache / Nginx with PHP support (XAMPP / WAMP / Laragon work perfectly)

2. Import the Database

Open **phpMyAdmin** (or MySQL CLI) and run:

```
SOURCE /path/to/booknest/database.sql;
```

Or paste the content of `database.sql` into the phpMyAdmin SQL tab.

3. Configure DB Credentials

Edit `includes/config.php` and update:

```
define('DB_HOST', 'localhost');
define('DB_NAME', 'booknest');
define('DB_USER', 'root');      // your MySQL username
define('DB_PASS', '');         // your MySQL password
```

4. Place Files in Web Root

Copy the `booknest/` folder to:

- **XAMPP:** `C:/xampp/htdocs/booknest/`
- **WAMP:** `C:/wamp64/www/booknest/`
- **Linux:** `/var/www/html/booknest/`

5. Open in Browser

```
http://localhost/booknest/
```



Features

Feature	Details
User Auth	Register, Login, Logout with bcrypt password hashing
Book Catalog	12 books loaded from MySQL, not hardcoded
Search & Filter	Live search by title/author, filter by genre, sort
Shopping Cart	Session-based for guests, DB-based for logged-in users; guest cart merges on login
Order Management	Place orders, reduce stock, view order history
Receipt System	Order confirmation with detailed receipt
Contact Form	Messages saved to database
Responsive UI	Works on mobile and desktop
Admin Panel	Basic admin functionality (expandable)

API Reference

Auth (`api/auth.php`)

Method	Action	Body	Description
POST	<code>?action=register</code>	<code>{full_name, email, password}</code>	Create account
POST	<code>?action=login</code>	<code>{email, password}</code>	Login user
POST	<code>?action=logout</code>	—	Logout
GET	<code>?action=me</code>	—	Session info

Books (`api/books.php`)

Method	Params	Description
GET	—	All books
GET	<code>?id=5</code>	Single book
GET	<code>?search=harry</code>	Search
GET	<code>?genre=Fiction</code>	Filter by genre
GET	<code>?sort=price-asc</code> / <code>price-desc</code> / <code>rating</code>	Sort

Cart (`api/cart.php`)

Method	Action	Body	Description
GET	—	—	Get cart
POST	<code>?action=add</code>	<code>{book_id, qty}</code>	Add to cart
POST	<code>?action=update</code>	<code>{book_id, qty}</code>	Change quantity
POST	<code>?action=remove</code>	<code>{book_id}</code>	Remove item
POST	<code>?action=clear</code>	—	Empty cart

Orders (`api/orders.php`) (*login required*)

Method	Params / Action	Description
POST	<code>?action=place</code>	Place order from cart
GET	—	Order history
GET	<code>?id=12</code>	Single order detail

Contact (`api/contact.php`)

Method	Body	Description
POST	<code>{name, email, subject, message}</code>	Submit contact form

Security Features

- Passwords hashed with **bcrypt** (`password_hash`)
- **PDO prepared statements** — no SQL injection possible
- Input validation on all API endpoints
- Session-based authentication (no JWTs needed for college project)
- CORS headers for local development

Database Tables

- `users` — registered accounts with roles
- `books` — product catalog with stock management
- `cart` — per-user cart items
- `orders` — placed orders with status tracking
- `order_items` — line items per order
- `contact_messages` — contact form submissions

Sample Books Included

The database comes pre-loaded with 12 popular books:

1. **The Alchemist** by Paulo Coelho - ₹299
2. **Atomic Habits** by James Clear - ₹499
3. **Harry Potter & the Sorcerer's Stone** by J.K. Rowling - ₹350
4. **A Brief History of Time** by Stephen Hawking - ₹399
5. **Sapiens** by Yuval Noah Harari - ₹599
6. **Clean Code** by Robert C. Martin - ₹799
7. **The Great Gatsby** by F. Scott Fitzgerald - ₹249
8. **Thinking, Fast and Slow** by Daniel Kahneman - ₹549
9. **Rich Dad Poor Dad** by Robert Kiyosaki - ₹349
10. **1984** by George Orwell - ₹199

11. **The Lean Startup** by Eric Ries - ₹449
 12. **Wings of Fire** by A.P.J. Abdul Kalam - ₹179
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How to Use

For New Users:

1. Click **Register** to create an account
2. Fill in your details (name, email, password 6+ chars)
3. Login with your credentials
4. Browse books and add to cart
5. Checkout to place orders

For Testing:

- **Admin Login:** `admin@booknest.com` / `booknest123`
 - Create regular user accounts through registration
 - Guest users can browse and add to cart (merges on login)
-

Technical Architecture

- **Backend:** PHP 7.4+ with PDO for database operations
 - **Database:** MySQL 5.7+ with proper foreign key relationships
 - **Frontend:** Vanilla JavaScript with fetch API for AJAX calls
 - **Styling:** Separated CSS file with clean, responsive design
 - **Session Management:** PHP sessions for authentication
 - **Security:** Bcrypt password hashing, prepared statements
-

Pages & Functionality

Public Pages

- **Home** - Welcome page with featured books
- **Catalog** - Full book collection with search/filter
- **Login** - User authentication
- **Register** - Create new account

User Pages (Login Required)

- **Cart** - Shopping cart management
- **Profile** - User account details
- **Order History** - View past orders

Admin Features

- User management
- Book inventory management
- Order tracking
- Contact message management

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